Continuous Control II: Max Entropy RL, SAC

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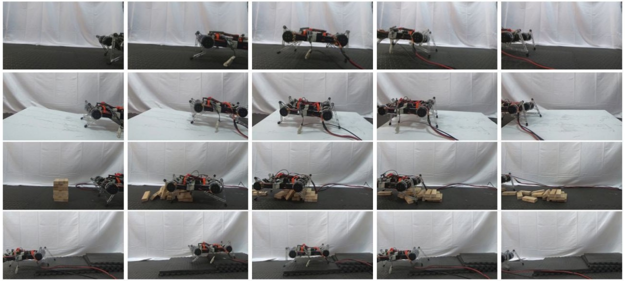

Soft Actor-Critic — *Haarnoja, Zhou, Abbeel, Levine* (2018)

Table of Contents

Soft Actor-Critic: **Recap (Day 5)**

- Day 5 gave us **off-policy continuous control**: DDPG [4] / TD3 [5] — a replay buffer, target networks, and a deterministic actor $\mu_\theta(s)$

Recap: Where Day 5 Left Us

- Day 5 gave us **off-policy continuous control**: DDPG [4] / TD3 [5] — a replay buffer, target networks, and a deterministic actor $\mu_\theta(s)$
- But DDPG is **fragile**:
  - deterministic actor $\Rightarrow$ *no native exploration* (hand-tuned OU/Gaussian noise bolted on)
  - brittle to hyperparameters and random seeds
  - mode collapse on multimodal Q-landscapes — commits to one of several equally good actions

Recap: What Today Delivers

- **Today — Soft Actor-Critic (SAC)** [1]: fix all three with a *stochastic* actor and an *entropy bonus* in the objective

$$J(\pi) = \sum_t \mathbb{E}_{(s_t, a_t) \sim \rho_\pi} \left[r(s_t, a_t) + \alpha \, \mathcal{H}(\pi(\cdot \mid s_t)) \right]$$

- Entropy drives *directed* exploration — no hand-tuned noise schedule; robust to seeds; captures multimodal optima
- **Everything from Day 5 carries over:** replay buffer, target networks, and TD3's clipped double-Q critic
- We close with **automatic temperature tuning** (SAC v2) — removing the last brittle hyperparameter

Scope & Notation

- Note: "Continuous Control II" is the *max-entropy* branch. PPO (Day 4) already handles continuous actions on-policy; DDPG/TD3 (Day 5) are the off-policy *deterministic* branch; SAC is the off-policy *stochastic, entropy-regularised* branch.
- **Notation:** π_ϕ stochastic actor, Q_θ soft critic, α entropy temperature, $\bar{\mathcal{H}}$ target entropy (used in SAC v2), $\mathcal{D}$ replay buffer — used consistently throughout.

Learning Outcomes

- Motivate the maximum-entropy RL objective and explain why the $-\log \pi$ bonus is principled (exploration, robustness, multimodality)
- State soft policy iteration — soft policy evaluation & improvement — and why it converges (Lemmas 1–2, Theorem 1)
- Derive the SAC critic and (reparameterised) actor objectives, including the squashed-Gaussian Jacobian correction
- Explain how SAC v2 $=$ SAC v1 $+$ TD3's clipped double-Q $+$ *automatic temperature tuning*, and what reward-scale sensitivity has to do with it
- Identify SAC's remaining limitations and place it in the algorithm family (REDQ / DroQ / TQC)

Main Problem: Sample Inefficiency

- Number of times the agent must interact with the environment in order to learn a task
- Good sample complexity is the first prerequisite for successful skill acquisition.
- Learning skills in the real world can take a substantial amount of time
  - can get damaged through trial and error

Main Problem: Sample Inefficiency

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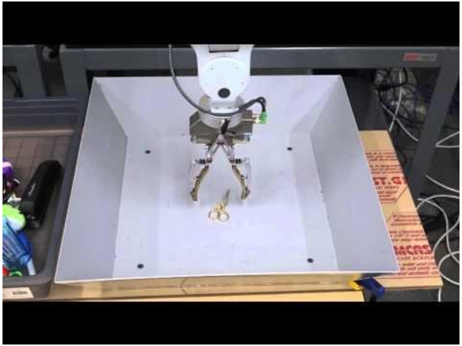
- "Learning Hand-Eye Coordination for Robotic Grasping with Deep Learning and Large-Scale Data Collection", Levine et al., 2016 [8]
  - 14 robot arms learning to grasp in parallel
  - objects started being picked up at around 20,000 grasps

https://spectrum.ieee.org/automaton/robotics/artificial-intelligence/google-large-scale-robotic-grasping-project

Main Problem: Sample Inefficiency

Observing the behavior of the robot after over 800,000 grasp attempts, which is equivalent to about 3000 robot-hours of practice, we can see the beginnings of intelligent reactive behaviors.



https://www.youtube.com/watch?v=cXaic_k80uM

- **Off-policy is the route to sample efficiency.** On-policy methods (PPO, A2C) discard data after each update; off-policy ones reuse it via a replay buffer $\mathcal{D}$ — SAC, like DDPG, is built on this.

Value-based (Day 2):

$$\text{Q-Learning:} \quad Q(s,a) \leftarrow Q(s,a) + \alpha\big(r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big)$$

$$\textbf{DQN } [6]\text{:} \quad \mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim U(\mathcal{D})}\Big[\big(r + \gamma \max_{a'} Q(s',a';\theta^-) - Q(s,a;\theta)\big)^2\Big]$$

- Replay buffer + target network θ^- stabilise off-policy TD — but $\max_a$ doesn't scale to continuous actions

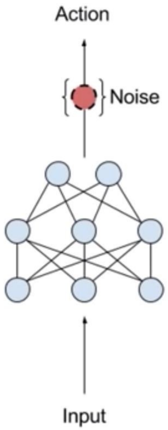
Actor-critic (Days 4–5):

$$\nabla \mathcal{J}(\theta) = \mathbb{E}_{\pi_\theta}[\,\nabla \ln \pi(a \mid s,\theta)\, Q_\pi(s,a)\,] \qquad \text{DDPG: } \nabla_\theta Q(s,\mu_\theta(s))$$

- DDPG/TD3 [4][5]: deterministic actor μ_θ — replay + targets reused, but exploration is just injected noise

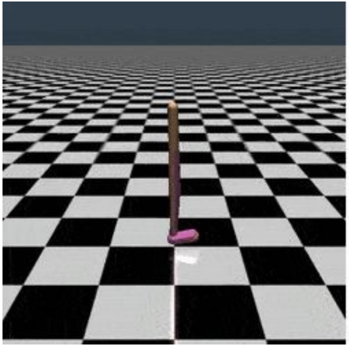
Prior Work: DDPG

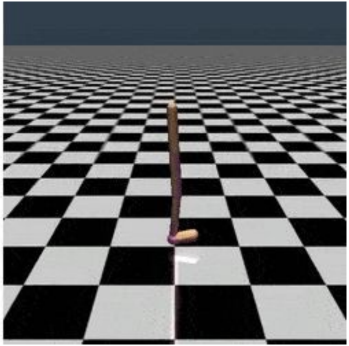
- DDPG = DQN + DPG (Lillicrap et al., 2015)
  - off-policy actor-critic method that learns a deterministic policy in continuous domain
  - exploration noise added to the deterministic policy when select action
  - difficult to stabilize and brittle to hyperparameters (Duan et al., 2016, Henderson et al., 2017)
  - unscalable to complex tasks with high dimensions (Gu et al., 2017)



https://www.youtube.com/watch?v=zR11FLZ-O9M

- Training is sensitive to randomness in the environment, initialization of the policy and the algorithm implementation

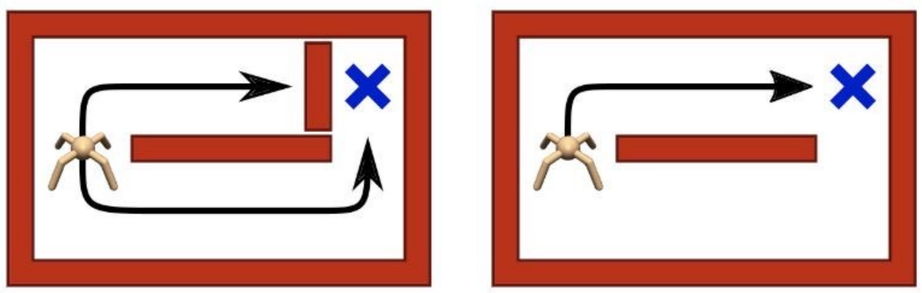




https://gym.openai.com/envs/Walker2d-v2/

Main Problems: Robustness

- Knowing only one way to act makes agents vulnerable to environmental changes that are common in the real-world



https://bair.berkeley.edu/blog/2017/10/06/soft-q-learning/

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Conventional RL Objective — Expected Reward

$$\sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s}_t,\mathbf{a}_t)]$$

Maximum Entropy RL Objective — Expected Reward + Entropy of Policy

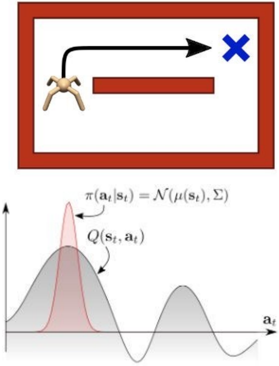
$$\sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s}_t,\mathbf{a}_t)+\alpha\,\mathcal{H}(\pi(\cdot\mid\mathbf{s}_t))]$$

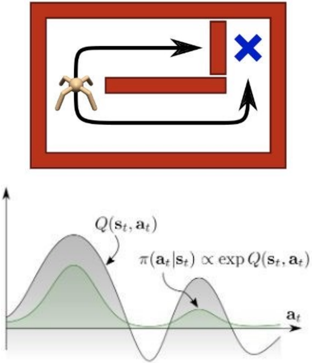
Entropy of a RV x

$$H(P)=\mathbb{E}_{x\sim P}[-\log P(x)]$$

Max Entropy RL

- MaxEnt RL agent can capture different modes of optimality to improve robustness against environmental changes





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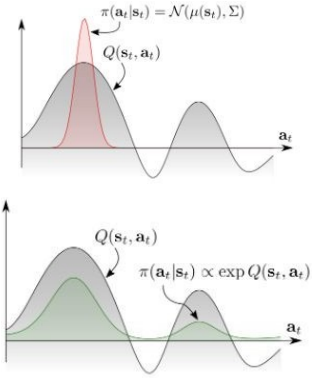


$$\min_\pi D_{\mathrm{KL}}\left(\pi(\cdot \mid s_0) \,\|\, \exp(Q(s_0, \cdot))/Z(s_0)\right)$$

$$= \min_\pi \mathbb{E}_\pi \left[\log \frac{\pi(a_0 \mid s_0)}{\exp(Q(s_0, a_0))}\right]$$

$$= \max_\pi \mathbb{E}_\pi [Q(s_0, a_0) - \log \pi(a_0 \mid s_0)]$$

(expand Q recursively over the horizon)

$$= \max_\pi \mathbb{E}_\pi \left[\sum_t r(s_t, a_t) + \mathcal{H}(\pi(\cdot \mid s_0)) \mid s_0\right]$$

$$= J_{\mathrm{MaxEnt}}(\pi(\cdot \mid s_0))$$

The partition function $Z(s_0) = \int \exp(Q(s_0, a))\, da$ is constant in π, so it drops from the arg min — the KL is well-defined against the *unnormalised* Boltzmann distribution. Z reappears explicitly on the SAC policy-improvement slide. The same objective also drops out of casting control as probabilistic inference — see Levine [3], *RL and Control as Probabilistic Inference* (2018); revisited under RLHF.

Prior Work: Soft Q-Learning

- **Soft Q-Learning** (Haarnoja et al., 2017) [7]
  - off-policy algorithms under MaxEnt RL objective
  - Learns Q^* directly
  - Instead of taking max action, sample the action
  - use approximate inference to sample
    - Stein variational gradient descent — evolves a set of particles to match the soft-Q Boltzmann distribution $\propto \exp\left(\frac{1}{\alpha} Q\right)$ without enumerating actions
  - not true actor-critic

Standard Q-learning (for contrast):

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left(R_{t+1} + \gamma \max_{a \in \mathcal{A}} Q(S_{t+1}, a) - Q(S_t, A_t)\right)$$

Soft versions (max $\rightarrow$ **soft-max via** $\log \int \exp$**):**

$$Q_{\text{soft}}(\mathbf{s}_t, \mathbf{a}_t) \leftarrow r_t + \gamma\, \mathbb{E}_{\mathbf{s}_{t+1} \sim p_{\mathbf{s}}}\left[V_{\text{soft}}(\mathbf{s}_{t+1})\right], \quad \forall \mathbf{s}_t, \mathbf{a}_t$$

$$V_{\text{soft}}(\mathbf{s}_t) \leftarrow \alpha \log \int_{\mathcal{A}} \exp\left(\tfrac{1}{\alpha} Q_{\text{soft}}(\mathbf{s}_t, \mathbf{a}')\right) d\mathbf{a}', \quad \forall \mathbf{s}_t$$

SAC: Contributions

- One of the most efficient model-free algorithms
  - SOTA off-policy
  - well suited for real world robotics learning
- Can learn stochastic policy on continuous action domain
- Robust to noise
- **Ingredients:**
  - Actor-critic architecture with separate policy and value function networks
  - Off-policy formulation to reuse previously collected data for efficiency
  - Entropy-constrained objective to encourage stability and exploration

- **policy evaluation:** compute value of π according to Max Entropy RL Objective
- modified Bellman backup operator $\mathcal{T}$:

$$\mathcal{T}^{\pi} Q(\mathbf{s}_t, \mathbf{a}_t) \triangleq r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p}\left[\boxed{V(\mathbf{s}_{t+1})}\right]$$

$$\boxed{V(\mathbf{s}_t)} = \mathbb{E}_{\mathbf{a}_t \sim \pi}\left[Q(\mathbf{s}_t, \mathbf{a}_t) - \boxed{\alpha \log \pi(\mathbf{a}_t \mid \mathbf{s}_t)}\right]$$

- **Lemma 1:** Contraction Mapping for Soft Bellman Updates

$$Q^{k+1} = \mathcal{T}^{\pi} Q^{k} \quad \text{converges to the soft Q-function of } \pi$$

Why: the standard Bellman contraction argument, unchanged — the extra entropy term $-\alpha \log \pi$ is bounded, so $\mathcal{T}^{\pi}$ is still a γ-contraction.

- **policy improvement:** update policy towards the exponential of the new soft Q-function
- modified Bellman backup operator $\mathcal{T}$:
  - choose tractable family of distributions big Π
  - choose KL divergence to project the improved policy into big Π

$$\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} D_{\text{KL}}\left(\pi'(\,\cdot \mid \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(\frac{1}{\alpha} Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot)\right)}{Z^{\pi_{\text{old}}}(\mathbf{s}_t)} \right)$$

- **Lemma 2:** the KL projection cannot decrease the soft value

$$Q^{\pi_{\text{new}}}(\mathbf{s}_t, \mathbf{a}_t) \geq Q^{\pi_{\text{old}}}(\mathbf{s}_t, \mathbf{a}_t) \quad \text{for any state action pair}$$

Why: π_{old} is feasible for the arg min, so the new policy's KL objective is no worse; expanding it and using KL ≥ 0 gives the monotone improvement.

Soft Policy Iteration

- soft policy iteration: soft policy evaluation $\longleftrightarrow$ soft policy improvement
- **Theorem 1:** Repeated application of soft policy evaluation and soft policy improvement from any policy $\pi \in \Pi$ converges to the optimal MaxEnt policy among all policies in Π

  Why: Lemma 2 makes $\{Q^{\pi^k}\}$ monotonically increasing; bounded reward + entropy keeps it bounded above; a monotone bounded sequence converges, and Lemma 1 says each evaluation step reaches its fixed point.

  - exact form applicable only in discrete case
  - need function approximation to represent Q-values in continuous domains
  - $\rightarrow$ Soft Actor-Critic (SAC)!

SAC — Parameterized soft Q-function

Symbol	Meaning	Example
$Q_\theta(\mathbf{s}_t, \mathbf{a}_t)$	parameterized soft Q-function	neural network
$\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)$	parameterized tractable policy	Gaussian, mean/cov. from a network
$J_Q(\theta),\ \hat{\nabla}_\theta J_Q(\theta)$	soft Q-function objective & its stochastic gradient	minimize soft Bellman residual
$J_\pi(\phi),\ \hat{\nabla}_\phi J_\pi(\phi)$	policy objective & its stochastic gradient	KL to $\exp(Q/\alpha)$, reparameterized

- **Critic — Soft Q-function:** minimize squared soft Bellman error; $\bar{\theta}$ is an EMA of the soft Q-weights (target network, from DQN)

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\mathcal{D}}\left[\tfrac{1}{2}\left(Q_\theta(\mathbf{s}_t,\mathbf{a}_t) - \left(r(\mathbf{s}_t,\mathbf{a}_t) + \gamma\,\mathbb{E}_{\mathbf{s}_{t+1}\sim p}[V_{\bar{\theta}}(\mathbf{s}_{t+1})]\right)\right)^2\right]$$

$$V_{\bar{\theta}}(\mathbf{s}_t) = \mathbb{E}_{\mathbf{a}_t\sim\pi_\phi}\left[\,Q_{\bar{\theta}}(\mathbf{s}_t,\mathbf{a}_t) - \alpha\log\pi_\phi(\mathbf{a}_t\mid\mathbf{s}_t)\right]$$

$$\hat{\nabla}_\theta J_Q(\theta) = \nabla_\theta Q_\theta(\mathbf{s}_t,\mathbf{a}_t)\left(\,Q_\theta(\mathbf{s}_t,\mathbf{a}_t) - \left(\,r(\mathbf{s}_t,\mathbf{a}_t) + \gamma\left[\,Q_{\bar{\theta}}(\mathbf{s}_{t+1},\mathbf{a}_{t+1}) - \alpha\log\pi_\phi(\mathbf{a}_{t+1}\mid\mathbf{s}_{t+1})\right]\right)\right)$$

SAC: Objectives and Optimization

- **Actor — Policy:** $\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} D_{\mathrm{KL}}\left(\pi'(\cdot \mid \mathbf{s}_t) \,\middle\|\, \exp\left(\frac{1}{\alpha} Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot)\right) / Z^{\pi_{\text{old}}}(\mathbf{s}_t)\right)$; multiply by α, drop Z:

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D},\, \mathbf{a}_t \sim \pi_\phi}\left[\alpha \log \pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t) - Q_\theta(\mathbf{s}_t, \mathbf{a}_t)\right]$$

- reparameterize: $\mathbf{a}_t = f_\phi(\epsilon_t; \mathbf{s}_t)$, noise $\epsilon_t \sim \mathcal{N}$ (spherical Gaussian)
  - **squashed Gaussian:** $\mathbf{a} = \tanh(\mu_\phi + \sigma_\phi \odot \epsilon)$ bounds actions; tanh adds a Jacobian term $-\sum_i \log\left(1 - \tanh^2(\cdot)\right)$ to $\log \pi_\phi$ (must-have; eval action $= \tanh(\mu_\phi)$)

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D},\, \epsilon_t \sim \mathcal{N}}\left[\alpha \log \pi_\phi(f_\phi(\epsilon_t; \mathbf{s}_t) \mid \mathbf{s}_t) - Q_\theta(\mathbf{s}_t, f_\phi(\epsilon_t; \mathbf{s}_t))\right]$$

$$\hat{\nabla}_\phi J_\pi(\phi) = \nabla_\phi \alpha \log(\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)) + \Big(\nabla_{\mathbf{a}_t} \alpha \log(\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)) - \nabla_{\mathbf{a}_t} Q(\mathbf{s}_t, \mathbf{a}_t)\Big) \nabla_\phi f_\phi(\epsilon_t; \mathbf{s}_t)$$

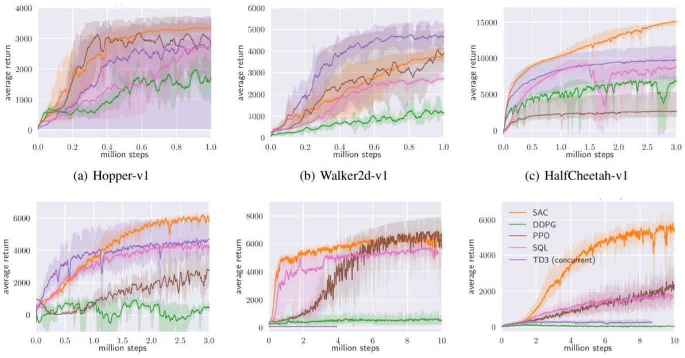
$\Rightarrow$ **Unbiased estimator extending DDPG-style policy gradients to any tractable stochastic policy**

SAC: Algorithm

Red = the two additions over SAC v1: *clipped double-Q* (TD3, Day 5) and *automatic temperature*. Shorthand $Q_\theta^{\min} \triangleq \min_{i=1,2} Q_{\theta_i}$.

Initialize $\theta_1, \theta_2, \phi, \alpha$; targets $\bar{\theta}_i \leftarrow \theta_i$; replay $\mathcal{D} \leftarrow \emptyset$
for *each iteration* **do**
  for *each environment step* **do**
    $\mathbf{a}_t \sim \pi_\phi(\cdot \mid \mathbf{s}_t)$; $\mathbf{s}_{t+1} \sim p$; $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1})\}$
  end
  for *each gradient step* **do**
    sample minibatch from $\mathcal{D}$; $\tilde{\mathbf{a}}_{t+1} \sim \pi_\phi(\cdot \mid \mathbf{s}_{t+1})$
    $y = r_t + \gamma\big(Q_{\bar{\theta}}^{\min}(\mathbf{s}_{t+1}, \tilde{\mathbf{a}}_{t+1}) - \alpha \log \pi_\phi(\tilde{\mathbf{a}}_{t+1} \mid \mathbf{s}_{t+1})\big)$
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i}\big[\frac{1}{2}(Q_{\theta_i}(\mathbf{s}_t, \mathbf{a}_t) - y)^2\big], \; i \in \{1, 2\}$
    $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$ (J_π critic term uses $Q_\theta^{\min}$)
    $\alpha \leftarrow \alpha - \lambda \hat{\nabla}_\alpha \big[-\alpha\big(\log \pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t) + \bar{\mathcal{H}}\big) \big]$
    $\bar{\theta}_i \leftarrow \tau\theta_i + (1-\tau)\bar{\theta}_i, \; i \in \{1, 2\}$
  end
end

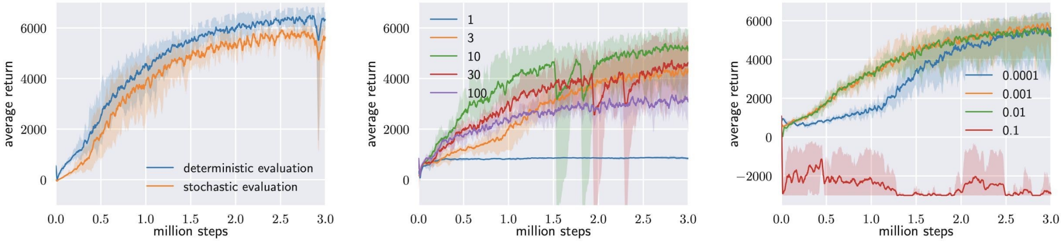
Haarnoja et al. 2018b, https://arxiv.org/abs/1812.05905



(a) Hopper-v1 (b) Walker2d-v1 (c) HalfCheetah-v1
(d) Ant-v1 (e) Humanoid-v1 (f) Humanoid (rllab)

Tasks: OpenAI gym continuous-control suite + the 21-dim rllab Humanoid (exceptionally hard for off-policy methods). **Baselines:** DDPG [4], SQL [7], PPO, TD3 (concurrent) [5].

⇒ SAC matches or beats DDPG/PPO/TD3/SQL on every benchmark, and is the *only* off-policy method to make real progress on Humanoid (rllab) — the headline win.



(a) Evaluation (b) Reward Scale (c) Target Smoothing Coefficient (τ)

Reward scale (left): performance is *dramatically* sensitive — too small $\rightarrow$ no exploration signal, too large $\rightarrow$ near-deterministic; the right scale is task-specific. **Target smoothing** τ (right): mild sensitivity, the usual stability/speed trade-off.

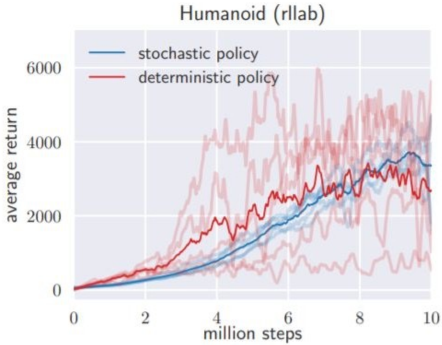
$\Rightarrow$ Reward scale is just a hand-tuned proxy for the entropy temperature α — *this* is the problem automatic temperature tuning solves.

https://arxiv.org/abs/1801.01290

Experimental Results: Ablation Study

- How does policy stochasticity and entropy maximization affect performance?
- Compared with a deterministic SAC variant (no entropy) that closely resembles DDPG

⇒ Stochastic+entropy is far more *stable across seeds*; the deterministic variant has DDPG-like high variance.



https://arxiv.org/abs/1801.01290

- SAC v1 fixed many of DDPG's issues, but inherited one: it is brittle to the temperature hyperparameter α that controls exploration (and α must be tuned per task)
    - $\rightarrow$ automatic temperature tuning (SAC v2) [2]

Contributions

- Adaptive temperature coefficient

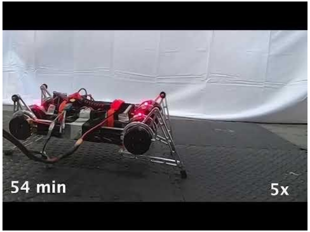
$$\sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}\left[r(\mathbf{s}_t,\mathbf{a}_t) + \underbrace{\alpha}_{\text{temperature}} \mathcal{H}(\pi(\cdot \mid \mathbf{s}_t))\right]$$

- Extend to real-world tasks such as locomotion for a quadrupedal robot and robotic manipulation with a dexterous hand

Real World Robots

Minitaur quadruped learning forward locomotion end-to-end with SAC on *real hardware* [2].

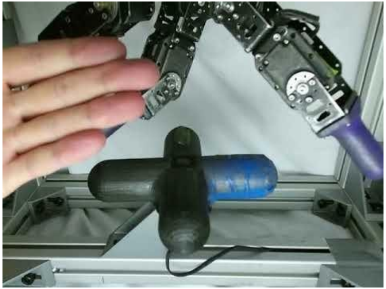
- No simulator, no curriculum
- Video overlays: "54 min" = elapsed real-world training; "5×" = playback speed
- Walks reliably within ≈2 h of real-world experience



https://arxiv.org/abs/1812.05905

Real World Robots

- **Task:** a multi-fingered hand opens a valve to a target angle, learning from raw joint sensors [2]
- End-to-end real-world learning in ≈20 hours wall-clock
- With valve angle exposed as input: **SAC 3 hours vs. PPO 7.4 hours** — the off-policy, entropy-regularised advantage shows clearly in sample efficiency



https://sites.google.com/view/sac-and-applications

Automatic Temperature Tuning

Constrained problem — maximise return subject to a minimum policy entropy $\bar{\mathcal{H}}$ (so we set a *target entropy* instead of a temperature):

$$\max_{\pi_{0:T}} \mathbb{E}_{\rho_\pi}\left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t)\right] \quad \text{s.t.} \quad \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t)\sim\rho_\pi}[-\log \pi_t(\mathbf{a}_t \mid \mathbf{s}_t)] \geq \bar{\mathcal{H}} \quad \forall t$$

Dual. Introduce a Lagrange multiplier $\alpha_t \geq 0$ per constraint (Levine [3]). Each inner problem is the usual MaxEnt objective with α_t as temperature; the multiplier itself minimises the dual:

$$\alpha_t^* = \arg\min_{\alpha_t \geq 0} \mathbb{E}_{\mathbf{a}_t \sim \pi_t^*}\left[-\alpha_t\left(\log \pi_t^*(\mathbf{a}_t \mid \mathbf{s}_t) + \bar{\mathcal{H}}\right)\right]$$

In practice — one SGD step on a single scalar, jointly with the actor/critic updates:

$$J(\alpha) = \mathbb{E}_{\mathbf{a}_t \sim \pi_\phi}\left[-\alpha\left(\log \pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t) + \bar{\mathcal{H}}\right)\right], \quad \alpha \leftarrow \alpha - \lambda \hat{\nabla}_\alpha J(\alpha)$$

Implementation Cheatsheet

Setting	**Default (Haarnoja 2018b)** [2]
Policy parameterization	Squashed Gaussian: $\mathbf{a} = \tanh(\mu_\phi + \sigma_\phi \odot \epsilon)$
Critic	Twin soft $Q_{\theta_1}, Q_{\theta_2}$; clipped double-$Q$ target
Target network update	Polyak: $\bar{\theta}_i \leftarrow \tau\theta_i + (1-\tau)\bar{\theta}_i, \quad \tau = 0.005$
Target entropy $\bar{\mathcal{H}}$	$\bar{\mathcal{H}} = -\dim(\mathcal{A})$ (one nat per action dim)
Optimizer	Adam, learning rate 3×10^{-4} for Q, π, and α
Discount γ	0.99
Replay buffer / batch	$\lvert\mathcal{D}\rvert = 10^6$, batch size 256
Reward scale (v1 only)	task-specific (replaced by auto-α in v2)

These are the defaults that ship with most SAC implementations (SB3, rlpyt, the authors' code). Tune only $\bar{\mathcal{H}}$ and network sizes per task.

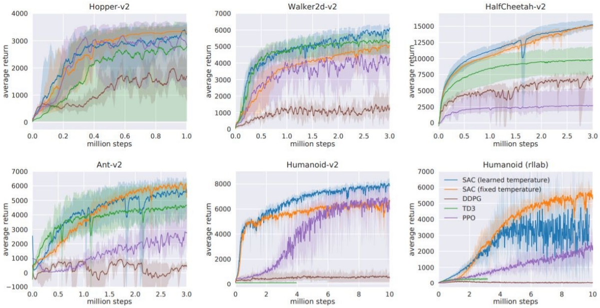


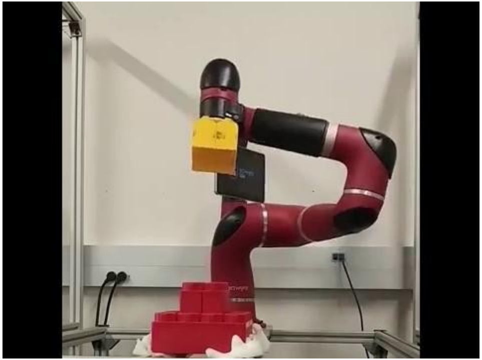
Figure 1: Training curves on continuous control benchmarks. Soft actor-critic (blue and yellow) performs consistently across all tasks and outperforming both on-policy and off-policy methods in the most challenging tasks.

Figure 1: Training curves on continuous control benchmarks. Soft actor-critic (blue and yellow) performs consistently across all tasks and outperforming both on-policy and off-policy methods in the most challenging tasks.

https://arxiv.org/abs/1801.01290

Task: Sawyer arm trained with SAC to stack a block.

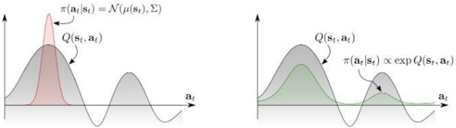
- Evaluated under *perturbations unseen at training time* (a human pushes the block / arm)
- The stochastic, entropy-regularised policy adapts to off-distribution states
- A deterministic policy would freeze on the trained trajectory

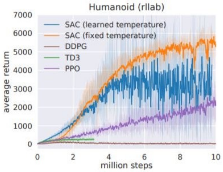


https://sites.google.com/view/sac-and-applications

Limitations / Open Issues

- Lack of experiments on hard-exploration problems (entropy is undirected — $\rightarrow$ Day 7)
- Approximating a multi-modal Boltzmann distribution with a *unimodal* Gaussian collapses modes
- High-variance gradients when using automatic temperature tuning on hard tasks (e.g. Humanoid rllab)





Soft Actor-Critic: **Summary**

The Algorithm Family Tree

DQN $+$ DPG $\to$ **DDPG** $\to$ $+$clipped double-Q $\to$ **TD3**
$\to$ $+$max-entropy / stochastic actor $\to$ **SAC v1** $\to$ $+$automatic temperature $\to$ **SAC v2**

Algorithm	**Off-policy?**	**Policy**	**Exploration**	**Stability**
DDPG	Yes (replay)	Deterministic	Injected noise	Brittle
TD3	Yes (replay)	Deterministic	Injected noise	Improved
SAC v1	Yes (replay)	Stochastic	Entropy bonus	Robust (α tuned)
SAC v2	Yes (replay)	Stochastic	Entropy bonus	Robust (auto α)

Summary: SAC

- An **off-policy maximum-entropy** deep RL algorithm: sample-efficient, scales to high-dimensional observation/action spaces, robust to seeds and noise
- **Theory:** convergence of soft policy iteration; the SAC objectives are its function-approximation instantiation
- **Algorithm:** soft critic (clipped double-Q) + reparameterised stochastic actor + automatic temperature — everything from Day 5 reused
- **Empirically:** outperforms DDPG, PPO, TD3 and Soft Q-Learning in optimality, sample complexity and robustness

Beyond SAC

- SAC (2018) is *still* the de facto default for continuous control in simulation and real-world robotics
- Direct descendants improved sample efficiency 5–10$\times$:

Method	**Idea**
TQC (2020) [9]	truncated quantile critics — controls overestimation
REDQ (2021) [10]	randomized ensemble of Q-functions, high update-to-data ratio
DroQ (2021) [11]	dropout-regularized critics — REDQ's efficiency, fewer nets

What's Next: Day 7

- SAC's entropy bonus *is* exploration via a stochastic policy — but it is undirected
- **Day 7 — Exploration:** directed strategies beyond entropy
  - UCB / optimism, count-based bonuses, curiosity / intrinsic motivation
  - the answer for hard-exploration, sparse-reward tasks SAC struggles on
- **Further out:**
  - **Day 8 — Offline RL:** on a fixed buffer the entropy bonus does *not* save you — SAC suffers DDPG's OOD-action extrapolation error (motivates BCQ/CQL)
  - **Day 9 — Real-world robot learning:** SAC is the workhorse for from-scratch robots (quadrupeds, dexterous hands) — revisited there
- Takeaway: entropy is *one* answer to exploration; Day 7 covers the rest

Soft Actor-Critic: **References**

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References

These slides have been adapted from

- Animesh Garg, CSC2621: Reinforcement Learning in Robotics, University of Toronto **[12]**
- Primary papers: Haarnoja 2018a **[1]** (SAC v1), Haarnoja 2018b **[2]** (SAC v2), Levine 2018 **[3]** (control as inference)